

Design of True Random Number Generator Based on Multi-Stage Feedback Ring Oscillator

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Abstract—True random number generators (TRNGs) play an important role in encryption systems. In this brief, a novel method of generating true random numbers on a field-programmable gate array (FPGA) is proposed based on the random jitter of a multi-stage feedback ring oscillator (MSFRO) as the entropy source. Based on the traditional ring oscillator, a multi-stage feedback structure is added to enlarge the range of clock jitter, and improve the frequency of clock sampling and the randomness of the entropy source. Different from the traditional clock sampling structure, we use the clock jitter signal generated by the MSFRO to sample the clock signal generated by the phase-locked loop (PLL) of the FPGA. The obtained output value is operated by XOR to reduce the deviation of the output value and improve its randomness. The TRNG is implemented in Xilinx Virtex-6 FPGA, which has low hardware resource consumption and high throughput. The entropy source classification, hardware resources and throughput are compared with those of existing TRNGs. The results showed that the proposed TRNG only consumed 24 LUTs and 2 DFFs. Compared with other TRNGs, this design has very low hardware resource consumption and its throughput is up to 290 Mbps. The random bit sequence generated by this TRNG passes the NIST SP800-22 test and the NIST SP800-90B test.

Index Terms—True random number generator, FPGA, multi-stage feedback, frequency accumulation.

I. INTRODUCTION

THE TRUE random number generator (TRNG) has an important role in many cryptographic systems, including password generation, authentication protocol, key generation, random padding and digital image encryption [1], [2].

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In addition, true random numbers have important applications in numerical calculations, statistical simulations, random sampling and quantum key distribution. The performance metrics of TRNGs include throughput, hardware resource consumption, and random number statistics. TRNGs strictly meet the statistical requirements, and unpredictability, and use random physical processes as entropy sources to generate random numbers. Entropy sources include thermal noise, the metastable state [3], clock jitter [4], chaos [5], and Magnetic Tunnel Junction (MTJ) [6], [7], [8]. If the randomness of the original random bitstream is not good, post-processing operations, such as Von Neumann correction or introducing a hash function, are needed to improve the randomness.

The entropy source of TRNGs based on FPGA design is generally the ring oscillator, DCM [9], the self-timed ring (STR) [4] and the meta-stability caused by the violation of the set time and hold time of the trigger [3]. TRNGs, which use clock jitter in RO circuit as an entropy source, can obtain good randomness under long time jitter accumulation, but the throughput will be reduced and the hardware resource consumption is large. It is mainly affected by the RO order that the output frequency decreases, resulting in lower throughput. The TRNG based on fast carry logic proposed can improve throughput, but in order to obtain more severe path delay, tedious layout wiring is needed to improve randomness. In TRNGs, which use PLL or digital clock manager (DCM) as the entropy source, the entropy source structure is simple, but the randomness is poor, and complex calculation is needed to find the appropriate parameters [9]. Therefore, in order to improve the throughput of TRNGs and reduce the consumption of hardware resources on FPGA. In this brief, we propose an entropy source structure that can improve the randomness of the entropy source. The TRNG composed of this entropy source has the following three advantages: The quality of the entropy source is good. The phase jitter of the clock can be increased, and the randomness of the entropy source can be improved in a short time by using the multi-stage feedback structure. The throughput is high, and the jitter signal frequency generated by the proposed entropy source is high. We use the jitter signal to sample the conventional clock signal, and the rate at which the random bit stream is generated is the frequency of the jitter signal. The sampling circuit is simple. And the consumption of hardware resources can be further reduced.

To demonstrate these advantages of the proposed structure, we implemented TRNG on several Virtex-6 and Spartan-6 FPGAs. The rest of this brief is structured as follows. In Section II, we introduce the generation principle of clock jitter and describe the relevant research. In Section III, we introduce our starting point for improving the randomness of clock jitter theoretically, and then systematically introduce the proposed

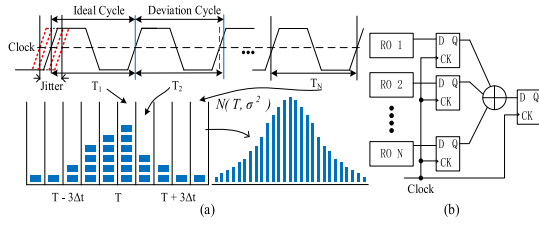


Fig. 1. Clock jitter following a Gaussian distribution (a) and TRNG based on parallel ROs (b).

TRNG. Section IV presents and discusses the randomness test. Lastly, conclusions are drawn in Section V.

II. RELATED RESEARCH

Clock jitter in the time domain and phase noise in the frequency domain are two ways that the clock signal is affected by noise. Ideally, a fixed pulse of frequency F should have a duration of exactly $T = 1/F$, with a jump edge at intervals of $T/2$. However, such signals do not exist. As shown in Fig. 1a, due to the thermal noise and interference of the circuit, the length of the signal cycle will always change to some extent, leading to uncertainty of the arrival time of the next jump edge. Therefore, in the time domain, the change of the clock frequency is shown as clock jitter, and in the frequency domain, it is shown as phase noise.

We take the cycle of the clock signal $T_1 \sim T_N$ as the sample, and Δt as the time interval. We conduct statistical analysis on the $T_1 \sim T_N$ sample. As shown in Fig. 1a, as Δt approaches infinity, the statistical distribution of random jitter follows a Gaussian distribution $N(T, \sigma^2)$, where T is the time point at which the ideal edge changes, and σ^2 is the variance of the jitter. The basic principle of generating random numbers by clock jitter is to obtain the uncertainty of rising or falling edges of the clock signal. Jitter is defined as the deviation between the timing event of a signal and its desired position. Total jitter can be divided into random jitter and deterministic jitter. Random jitter is regarded as an entropy source, and is mainly caused by noise or other interference in the system. If we sample the data within the range of the jitter, then we can obtain a random number. In TRNG designs based on FPGA, the clock jitter of the RO is commonly used as the entropy source. However, the jitter range is very narrow, and difficult to extract. In order to expand the range of jitter, the RO needs to accumulate clock jitter for a long time or increase the order, but then the frequency of the RO will decrease. Thus the throughput of TRNGs based on RO design is generally low. In [10], multiple parallel ROs are used to generate random bit streams, as shown in Fig. 1b. The purpose is to improve the quality and throughput of the entropy source, but this will increase the complexity of circuit design and consume a lot of hardware resources. We can improve the throughput and reduce the consumption of hardware resources by improving the entropy source structure without increasing the complexity of the circuit.

III. MULTI-STAGE FEEDBACK RING OSCILLATOR

In digital circuits, jitter will appear at the rising or falling edge of the RO clock and propagate and accumulate in the RO due to semiconductor noise temperature change crosstalk and propagation delay. The period of the RO is approximately $T = 2T_{delay}N$, where T_{delay} is the delay of a single inverter

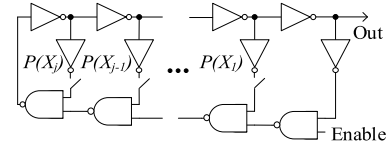


Fig. 2. Schematic diagram of MSFRO.

and N is the number of inverters in the RO. To increase the frequency of the RO, it is necessary to reduce T_{delay} and the number of inverters N . Meanwhile, to improve the RO's clock jitter, it is necessary to increase the number of inverters N ; this will also reduce the frequency of RO. In [11], the relation between the phase noise of the ring oscillator and the order of the oscillator is proposed. Its expression is

$$L(f)_{min} = \frac{8}{3\eta} N \frac{KT}{p} \left(\frac{V_{DD}}{V} + \frac{V_{DD}}{IR} \right) \left(\frac{f_0}{f} \right)^2 \quad (1)$$

where K is the Boltzmann constant; T is the absolute temperature; η , V_{DD} , V , R , and I are constants; f_0 is the frequency of the RO; f is the offset frequency; p is the power consumption of the ring oscillator; and N is the order of the ring oscillator. When the frequency is not fixed, increasing N will reduce the frequency f_0 , which will optimize the phase noise and reduce the throughput. The relation between phase noise in the frequency domain and mean jitter in the time domain is

$$\sigma_{RMS} = \frac{\Delta\phi}{2\pi f_0}, \Delta\phi = \sqrt{2 \int_{f_1}^{f_2} L(f) df} \quad (2)$$

from equation (2), we can see that the mean jitter is positively correlated with the phase noise. In addition, when the signal frequency f_0 is increased, the mean jitter σ_{RMS} becomes smaller and the clock jitter can be optimized. Thus, the randomness of the entropy source becomes worse. In order to solve this kind of contradiction, we add a multi-stage feedback structure to the RO circuit; this is equivalent to increasing the order of the RO. The feedback structure is a single inverter. Our purpose is to combine each feedback structure and some of the RO circuit inverters to form a new RO that can operate independently, so that the whole circuit can continuously oscillate. The feedback structure can increase the phase noise and thus the randomness of the entropy source can be improved. As shown in Fig. 2, the proposed MSFRO is a multi-stage feedback structure. The number of feedback structures is odd, because when the feedback structures appear in pairs, they will be highly correlated due to coupling, which will greatly reduce the randomness of the output bitstream of TRNG [12].

As shown in Fig. 3a, at the j -th stage of feedback, when $A = 1$, $B = 0$ is obtained after passing through the inverter of the feedback part. C is the value propagated from the previous order. Then B and $C=0$ are operated by NAND to get $D = 1$. After D passes through the inverter, the value of A changes from 1 to 0, forming an oscillation ring and shortening the oscillation period. Similarly, oscillating rings can also be formed when $A = 0$. As shown in Fig. 3b, at the i -th stage of feedback, when $A = 1$, $B = 0$ is obtained after the feedback. Then B and $C = 0$ are operated by NAND to get one value. After passing through m inverters, this value yields $D = 1/0$, where $m = j - i + 1$. After D passes through m inverters, the value changes from 1 to 0. Through the feedback part of the circuit, the oscillation period can be effectively shortened and the oscillation frequency can be increased.

In this brief, we propose a method wherein the frequency of the structure can be higher than that of the traditional RO. This method is called frequency accumulation. First, we take the Fourier transform of the square wave and obtain following expression:

$$f(t) = \frac{2E}{\pi} [\sin(\omega_0 t) + \frac{1}{3} \sin(3\omega_0 t) + \cdots + \frac{1}{n} \sin(n\omega_0 t)] \quad (3)$$

E is the amplitude of the square wave. A new RO can be constructed internally by introducing multiple stages of feedback into the RO ring. Each new RO outputs a square wave at the same time, and these square waves are accumulated to construct a new square wave. We suppose that the Fourier transform of each square wave is expanded as $f_1(t)$, $f_2(t)$, $f_j(t)$. The sum of the expansions of these square waves gives the following expression:

$$\begin{aligned} F(t) &= f_1(t) + f_2(t) + f_3(t) + \cdots + f_j(t) \\ &= \frac{2E}{\pi} [\sin(\omega_1 t) + \sin(\omega_2 t) + \cdots + \sin(\omega_j t)] \\ &\quad + \cdots + \frac{1}{3} [\sin(3\omega_1 t) + \sin(3\omega_2 t) + \cdots + \sin(3\omega_j t)] \\ &\quad + \cdots + \frac{1}{n} [\sin(n\omega_1 t) + \sin(n\omega_2 t) + \cdots + \sin(n\omega_j t)] \end{aligned} \quad (4)$$

Equation (4) can be approximately equal to

$$F(t) = \frac{2E}{\pi} \left[\sin(\omega t) + \frac{1}{3} \sin(3\omega t) + \cdots + \frac{1}{n} \sin(n\omega t) \right] \quad (5)$$

where $\omega = [\omega_1, \omega_2, \omega_3, \dots, \omega_j]$. It can be seen that equation (5) should be a square wave. Moreover, the frequency of the square wave is $f \geq \max[\frac{\omega_1}{2\pi}, \frac{\omega_2}{2\pi}, \dots, \frac{\omega_j}{2\pi}]$. The MSFRO structure is defined using the following feedback polynomial:

$$f(x) = \sum_{i=1}^j P(x) \cdot x^i + 1 \quad (6)$$

where $j = \frac{n-3}{2}$, $n > 3$ and $n = 2k + 1$, $k > 1$ and $k \in \mathbb{N}^*$. j is the number of feedback partial inverters, n is the number of inverters in the traditional RO, $P(x)$ is the feedback polynomial coefficient, and $P(x)$ is a constant expression, that is, $P(x) = 1$ or $P(x) = 0$. $P(x) = 1$ means there are feedback connections in the loop, while $P(x) = 0$ means there are no feedback connections in the loop. If all the coefficients of the feedback polynomial $P(x)$ are 0, the feedback polynomial becomes $f(x) = 1$, thus indicating that there is no feedback in the loop. We experimented on the 9th-order MSFRO with the four different modes. Their formulas are $f(x) = x + 1$, $f(x) = x^2 + 1$, $f(x) = x^3 + 1$, $f(x) = x^3 + x^2 + x + 1$, respectively. As shown in Fig. 4, our TRNG structure uses two MSFROs as the jitter signal source. The PLL generates two normal clocks. The output of PLL is used as the data input of flip-flop, and the output of MSFRO is used as the clock signal input of flip-flop. Each time the rising edge of the MSFRO output signal comes, the D flip-flop will sample the PLL output signal to generate one random bit. The phase jitter range caused by noise is very small, and many definite values will be sampled when using the trigger to sample, which will reduce the randomness of random numbers. Therefore, the proposed TRNG uses two MSFRO as the entropy source, and directly obtains random numbers by XOR output after extracting their randomness.

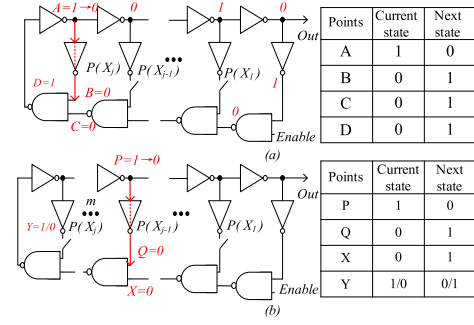


Fig. 3. The diagram of the j -th (a) and i -th (b) stage feedback.

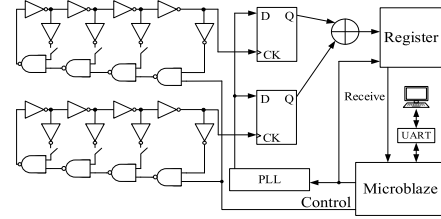


Fig. 4. TRNG design block diagram.

TABLE I
RESULTS OF NIST SP 800-22 TEST

NIST SP800-22	chip#1		chip#2		chip#3	
	P-value	Prop	P-value	Prop	P-value	Prop
Approx-Entropy	0.4460	10/10	0.6202	10/10	0.7992	10/10
Block-Freq.	0.9616	10/10	0.8427	10/10	0.5280	8/10
Cumsum	0.5355	10/10	0.4349	10/10	0.5474	10/10
FFT	0.2119	8/10	0.7620	9/10	0.8688	10/10
Frequency	0.4889	10/10	0.3391	10/10	0.7278	10/10
Line-Complex	0.9097	10/10	0.4254	10/10	0.8549	10/10
Long-Run	0.5486	9/10	0.3782	10/10	0.9151	10/10
Nonoverlapping	0.4826	10/10	0.5315	9/10	0.4835	10/10
Overlapping	0.8757	10/10	0.5621	10/10	0.6728	9/10
Rand-Excursion	0.4760	9/10	0.4650	10/10	0.3637	10/10
Rand-Variant	0.4006	10/10	0.5500	10/10	0.4740	10/10
Rank	0.6433	10/10	0.1978	10/10	0.3007	10/10
Runs	0.1431	10/10	0.7332	10/10	0.9316	10/10
Serial	0.6483	10/10	0.2528	9/10	0.1925	10/10
Universal	0.1426	10/10	0.8515	10/10	0.3302	10/10

*For the non-overlapping template, random excursions variant and random excursions, the p-value is the average of the p-values of all subtests

IV. EXPERIMENTAL RESULTS

A. NIST SP 800-22 Test

This test verifies the randomness of the sequence generated by the proposed TRNG under standard operating conditions (25 °C, 1.0 V). In order to ensure the accuracy of the experimental data and avoid the contingency of the experimental data of a single development board, the experiment was carried out on three different Virtex-6 FPGAs development boards. Under different working conditions, one million bits are generated continuously to test the randomness. The Prop is the random pass rate in the ten tests. The test results are shown in Table I, the random bitstream can pass each random test with a high P-value. The low P-value is probably due to the influence of device differences and different board layouts. However, all passed the test, thus indicating that the TRNG can generate true random numbers.

B. NIST SP 800-90B Test

The NIST SP800-90B test is more complex and rigorous than existing entropy estimation methods. Table II shows the results of the IID test, the chi square test and the longest repeat substring length test (LRS test). The experimental results show

TABLE II
RESULTS OF NIST SP 800-90B TEST

Test	Result		
	C[i][0]	C[i][1]	IID
Excursion	2872	0	pass
NumDirectionalRuns	1041	13	pass
LenDirectionalRuns	843	1763	pass
NumIncreasesDecreases	8859	15	pass
NumRunsMedian	4781	12	pass
LenRunsMedian	3748	2470	pass
AvgCollision	7673	1	pass
MaxCollision	9164	462	pass
Permutation tests	Peri-1	7618	27
	Peri-2	5450	17
	Peri-8	4692	29
	Peri-16	7528	27
	Peri-32	4317	17
Covariance	Cov-1	477	2
	Cov-2	3371	8
	Cov-8	5573	7
	Cov-16	5237	2
	Cov-32	3227	4
chi square test	Compression	9266	38
	Independence		pass
LRS test	Goodness-of-fit		pass
	Min-entropy		0.985281

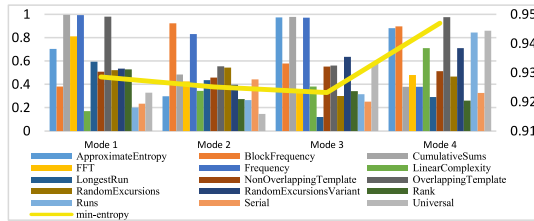


Fig. 5. NIST test results in four different modes.

that the minimum entropy of the bitstream sequence generated by the TRNG is 0.985281, and it passes the IID test of the NIST SP800-90B test. Therefore, the bitstream sequence generated by the TRNG is a random number.

C. NIST Test Results in Four Different Modes

In order to further verify the reliability and rationality of the structure, we conducted experiments on the four modes. The generated random bitstream was tested by NIST SP800-22 and SP800-90B. As shown in Fig. 5, all bitstreams pass the NIST SP800-22 test, indicating that they are true random numbers. All the data were re-tested by NIST SP800-90B to further determine the entropy source quality of the obtained random numbers. The broken line in Fig. 6 represents the minimum entropy values for the four modes. The minimum entropy of mode 4 is 0.946909, higher than that of the other three modes. This occurs because mode 4 has three feedback structures, which can significantly increase the phase noise in the circuit, increase the quality of the entropy source.

D. Deviation Test and Autocorrelation Test

Figure 6a presents images generated by three sets of one million contiguous bits. We can clearly see that the distribution of black and white pixels in the images generated in this study is very uniform, and thus the generated data have no offset and good randomness. To test the autocorrelation of random sequence is to verify the randomness of the sequence. The degree of correlation is expressed by the correlation coefficient. According to the statistical index designed by Karl Pearson, the correlation coefficient is less than 0.3 and the correlation can be regarded as irrelevant. The autocorrelation

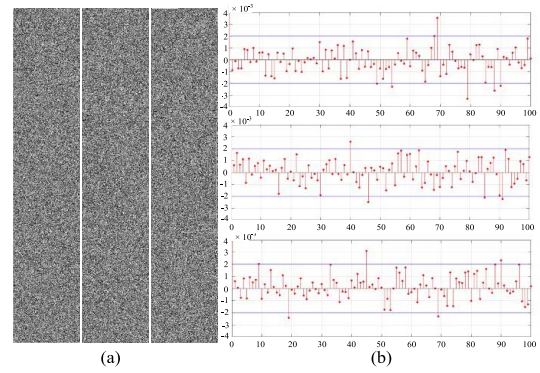


Fig. 6. Three groups of bitstream images (a) and autocorrelation test results (b).

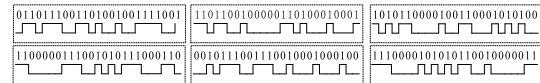


Fig. 7. Results of six restart tests.

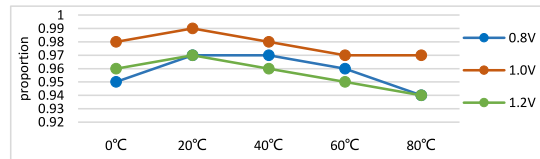


Fig. 8. Results of voltages and temperatures test.

test results of the three groups of data measured by the structure of mode 4 are presented in Fig. 6b. As can be seen from the figure, the correlation coefficient is below 0.3 for all groups of data. Therefore, the random sequence generated by the proposed TRNG does not have autocorrelation.

E. Restart Test

In the restart test [13], we plotted the data for the first several sampling bits of six restart tests. As shown in Fig. 7, if the random sequence shows a different graph, then the data obtained are a truly random sequence. The experimental results showed that each test yielded a different random sequence. Therefore, the random sequence generated by the TRNG designed in this brief does not have repeatability and similarity, and is a true random sequence.

F. Voltages and Temperatures Test

We evaluate the performance of TRNG at different voltages and temperatures on Xilinx Virtex-6 FPGA. Some experiments are conducted under different environments (temperatures: 0 °C ~ 80 °C) and voltages (0.8 V~1.2 V). As shown in Fig. 8, at different temperatures and voltages, 100 sets of experiments were conducted, each of which collected one million bits of bitstream data. The NIST test was performed on each set of measured data to determine the percentage that passed the test. As can be seen from the Fig. 8, at 1.0 V and 20 °C, the pass rate is the best, so the randomness is the best. When the voltage changes, the pass rate decreases and the randomness decreases. As the temperature rises from 20 °C, the pass rate gradually decreases and the randomness gradually becomes worse.

TABLE III
COMPARISON OF THROUGHPUT AND HARDWARE
RESOURCE CONSUMPTION

Design	Entropy source	Resource	Rate(Mbps)
[1]	RRAM	--	6
[4]	STR	320 LUTs 320 DFFs	200
[14]	STR	32 LUTs 48 DFFs	4
[15]	RO	83 LUTs 26 DFFs	100
[10]	RO	526 LUTs 177 DFFs	6
[13]	RO	64 LUTs	27
[16]	RO	75 LUTs 419 DFFs	120
[17]	FIGARO	866 LUTs	6.25
[18]	GARO	50 LUTs 79 DFFs	280
[19]	CMOS	--	192.3
This work	MSFRO	24 LUTs 2 DFFs	150(Spartan-6)
This work	MSFRO	24 LUTs 2 DFFs	290(Virtex-6)

TABLE IV
THE POWER OF FOUR MODES ON VIRTEX-6 FPGA

Mode	1	2	3	4
Power(w)	3.687	3.687	3.687	3.703

G. Comparison With Other TRNG Based on FPGA

We use the experimental results of mode 4 to compare with other TRNGs, and the results are shown in Table III. The structure in [1] adopts RRAM as the entropy source, and has a low throughput of 6 Mbps. The structure in [14] adopts STR as the entropy source, and has a low throughput of 4 Mbps. In [4], the structure adopts STR as the entropy source to sample multiple DFFs in parallel, has a high throughput of 200 Mbps. However, the hardware resource consumption is very serious. This is very unfavorable for FPGA with very limited resources. The structures in [10], [14], [15] and [16] all use the RO as the entropy source. The throughputs of [15] and [16] are 100 Mbps and 120 Mbps respectively, which are significantly higher than those of [10] and [13] but lower than that of our proposed TRNG (which is 150 Mbps). Moreover, the hardware resource consumptions of [15] and [16] are very large. The TRNG in [13] has low hardware resource consumption and low throughput. The TRNG in [10] not only consumes a lot of hardware resources but also has a low throughput. The structure in [17] adopts FIGARO as the entropy source, and has a low throughput of only 6.25 Mbps. The structure in [18] adopts GARO as the entropy source, has a throughput of 280 Mbps, and the structure in [19] adopts CMOS as the entropy source, has a throughput of 192.3 Mbps, but lower than that of our proposed TRNG (which is 290 Mbps). Overall, compared with other architectures, the TRNG proposed in this brief has less hardware resource consumption and higher throughput. We set the frequency range of the external periodic signal as 285~295 MHz for the injection locking test. A total of 10 groups of random bitstream data were tested, and the randomness pass rate was 70%. Furthermore, we used Xilinx XPower Analyzer in ISE to analyze the power consumption of the four modes. As shown in Table IV, the power consumption of mode 4 is relatively high due to the existence of multiple feedback structures.

V. CONCLUSION

We use the clock jitter of MSFRO proposed in this brief to generate randomness. Our TRNG can obtain higher throughput and lower hardware resource overhead than existing TRNGs, and without complex extraction structure. Through experimental analysis, it is concluded that the designed TRNG has good performance and all pass the randomness tests. The proposed TRNG has a throughput of 150 Mbps on Spartan-6 FPGA and 290 Mbps on Virtex-6 FPGA, and a hardware resource

overhead of only 24 LUTs and 2 DFFs. Hence, we save significant hardware resources and provide a more compact TRNG design.

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